Song Recommendation System

Project Report

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VIT BHOPAL

Python | Pandas | Machine Learning

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1 Project Overview

Title: Music Recommendation System using Python, Pandas, and Machine Learning

This project implements a **Song Recommendation System** that:

- Suggests **top 5 songs** based on user input (language/genre)
- Uses **Pandas** for data processing
- Applies **MinMaxScaler** for feature normalization
- Ranks songs using **danceability** + **energy** score

2 Objectives

1. Build functional recommendation engine
2. Implement intelligent song ranking
3. Demonstrate data science concepts
4. Create user-friendly interface

3 Technologies

Technology	Purpose
Python 3.x	Core programming
Pandas	Data manipulation
Scikit-learn	Feature scaling
CSV	Data storage

4 System Flow

```
[User Input: "hindi"]
       ↓
[Load: my_songs.csv]
       ↓
[Filter: genre contains "hindi"]
       ↓
[Score: danceability + energy]
       ↓
[Sort: Top 5 highest scores]
       ↓
[Display: track_name, artist, genre]
```

5 Core Implementation

5.1 Data Loading

```
try:
    df = pd.read_csv('my_songs.csv')
except FileNotFoundError:
    return "Please run data_gen.py first!"
```

Listing 1: Data Loading

5.2 Ranking Algorithm

```
scaler = MinMaxScaler()
results['rank_score'] = results[['danceability', 'energy']].sum(axis=1)
top_5 = results.sort_values(by='rank_score', descending=True).head(5)
```

Listing 2: Ranking Logic

Scoring Formula: $ rank_score = danceability + energy $

- Range: [0, 2]
- Higher score = Better recommendation

6 Sample Execution

Input:

`Enter a Language or Genre: hindi`

Output:

```
Top Recommendations:
track_name         track_artist     genre
Dil Diyan Gallan   Atif Aslam       Hindi
Raabta            Asees Kaur       Hindi
Gallan Goodiyaan Yashita Sharma   Hindi
```

7 Algorithm Analysis

Operation	Time Complexity	Space
Loading	O(n)	O(n)
Filtering	O(n)	O(k)
Sorting	O(k log k)	O(1)

8 Key Features

- **Smart Search**: Keyword matching in genres
- **ML Integration**: Feature scaling & ranking

- **Error Handling**: File/empty result cases
- **Efficient**: Fast processing even for large datasets

9 Future Enhancements

- GUI interface (Tkinter/Streamlit)
- Collaborative filtering
- Real-time Spotify API
- User preference profiles
- Advanced ML models

10 Conclusion

This project successfully demonstrates:

- Data science pipeline implementation
- Custom algorithm design
- Professional Python coding practices
- Real-world ML application

The system provides a solid foundation for advanced recommendation engines.